

University of Information Technology & Sciences (UITs)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Mid Term Examination, Spring-2024

Course Title: Data Structures and Algorithms I

Course Code: CSE 211

Marks: 20

Time: 1 (One) Hour

(Answer all questions)

1. a) Briefly explain the classification of data structure with example. [03]
b) Let A be an $n \times n$ matrix. Write a module which [05]
 - i. Find the number NUM of nonzero elements in A.
 - ii. Find the product PROD of diagonal elements.
- c) Suppose you are designing a memory layout for a 2D array in row-major and column major order. The base address of the array is 2000, and the array dimensions are 4×3 , with each element being of type float (4 bytes). Calculate the memory location of the element at location (2, 1). [02]
2. a) Define circular linked list with proper figure. Build an algorithm to insert a node into linked list at given position. [04]
b) The memory representation of a linked list in memory is as follows (Figure 01): [03]

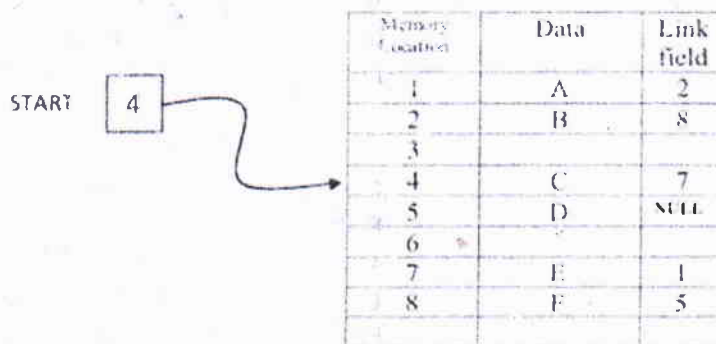


Figure 01: Memory representation of linked list

- i. Find the sequence of characters in the list.
- ii. Suppose F and then C are deleted from the list. Draw the final structure.
- iii. Suppose F and then C are deleted from the list and then G (You may use the empty memory locations) is inserted at the beginning of the list. Draw the final structure.
- c) Compare between array and linked list in the context of insertion, accessing, memory allocation and memory utilization. [03]

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Time: 03(three) Hours

(Answer all questions)

1. a) Given the following sequence of letters and asterisks: [03]

THIS*IS*A*TE*STSTR**IN*G****

Consider the stack data structure, supporting two operations push and pop. Suppose that for the above sequence, each letter (such as T) corresponds to a push of that letter onto the stack and each asterisk (*) corresponds to a pop operation on the stack. Show the sequence of values returned by the pop operations.

- b) Consider the following postfix expression: [03]

2 5 3 6 + * * 5 / 2 -

Evaluate this postfix expression, demonstrate step-by-step how you would compute the result. Describe each operation performed on the stack, including pushing operands and applying operators. Begin with an empty stack and show the sequence of operations until you obtain the final result of the expression.

- c) Using a stack-based algorithm, convert the following infix expression to postfix notation. [04]

(((P + Q) * (R + S)) / T) + (A * (B + C))

2. a) You have a sorted list of member ages: [04]

2, 5, 8, 12, 16, 23, 38, 56, 72, and 91

Using a appropriate search algorithm, demonstrate step-by-step how you would determine if a person of age 23 exists within this list.

- b) Define adaptive and non-adaptive sorting. Apply the bubble sort algorithm to sort the following list of alphabets "PARADISE". (Show all steps) [06]

3. a) Define Full Binary Tree. Construct a Binary Search Tree (BST) by inserting the following sequence of numbers: [03]

18, 14, 2, 10, 26, 51, and 11

- b) Consider the following letter of frequencies: [04]

a	b	c	d	e	f
22	19	5	11	6	7

Use the given frequencies to build a Huffman tree. There isn't just one solution, so your answer may be different from others.

- c) Given two integer arrays inorder and preorder where inorder is the inorder traversal of a binary tree and preorder is the preorder traversal of the same tree, construct the binary tree. [03]

Inorder = [4, 10, 12, 15, 18, 22, 24, 25, 31, 35, 44, 50, 66, 70, 90]

Preorder = [25, 15, 10, 4, 12, 22, 18, 24, 50, 35, 31, 44, 70, 66, 90]

4. a) Consider a graph with the following adjacency list: [03]

A : [B, D, E]

B: [A]

C: [E]

D: [A, E]

E: [A, D, C]

i. Write down the adjacency matrix for this graph.

ii. Draw this graph.

- b) Apply Prim's algorithm to find a minimum spanning tree from the following graph where the starting node is A (Figure 01). [04]

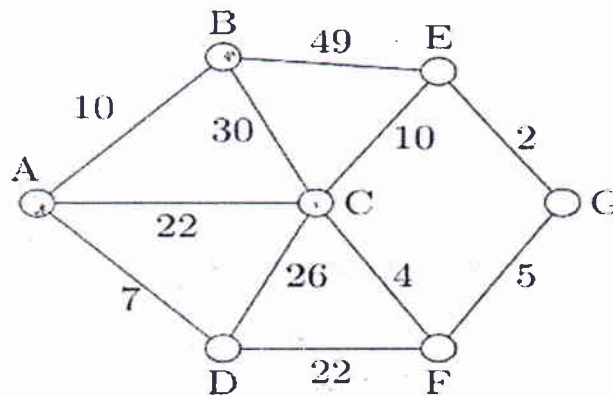


Figure 01: Weighted undirected graph

- c) Suppose the neighbors of a given vertex are examined in alphabetic order, and that vertex A is the starting point. Show Depth First Search (DFS) tree of the following graph. [03]

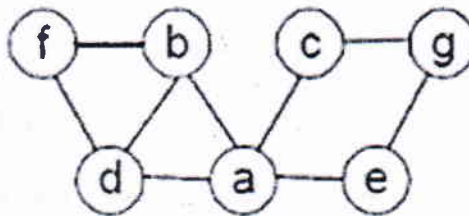


Figure 02: Undirected graph

5. a) List several real-life scenarios and applications where the use of stack and queue data structures are prevalent. [03]
- b) Explain how the input and output operations of a priority queue differ from those of a normal queue. [04]

- c) Consider the following dequeue of characters where DEQUE is a circular array which is allocated six memory cells: [03]

LEFT = 2 RIGHT = 4

DEQUE: _____, A, C, D, _____, _____

Describe the dequeue while the following operations take place.

- i. F is added to the right of the deque.
- ii. Two letters on the right are deleted.
- iii. K, L, and M are added to the left of the dequeue.
- iv. One letter on the left is deleted.
- v. R is added to the left of the dequeue.
- vi. S is added to the right of the dequeue.